

Smart Marking Setup Guide

From an empty Railway account to students reading their feedback on a phone — every screen is the real app.

VERSION 2 · SEPTEMBER 2026 ABOUT 20 MINUTES NO CODING NEEDED

STEP 01 Before you start

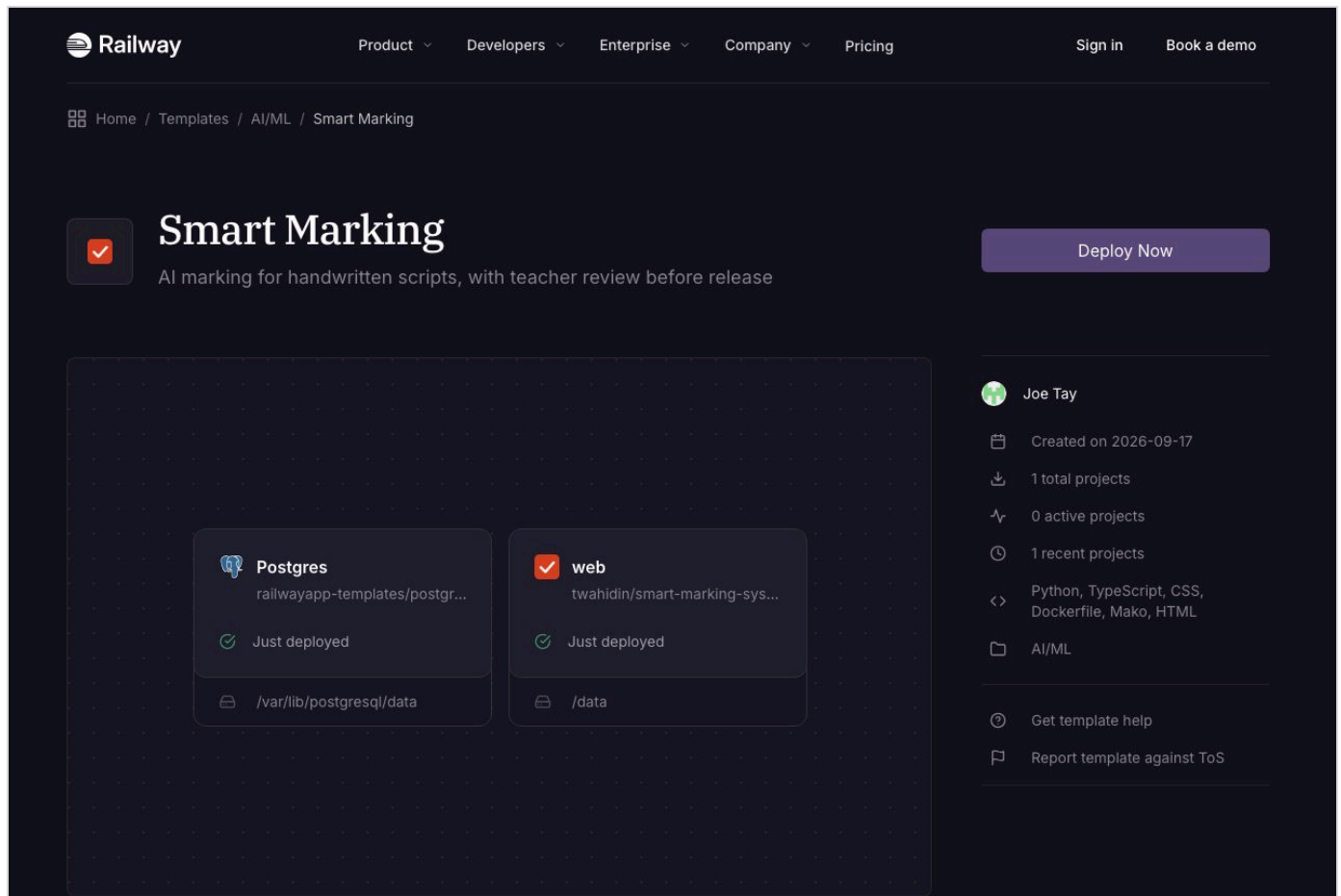
You need three things: a **Railway account** (railway.com — the Hobby plan is enough for a school department), an **API key for one AI provider** that can read images, and about twenty minutes. Free keys that work: Google Gemini's free tier (aistudio.google.com/apikey) and OpenRouter's `:free` models. Paid options: TokenRouter, OpenRouter, OpenAI, Anthropic, Moonshot (Kimi) and Qwen.

Everything — scripts, marks, classes, your key — lives inside your own Railway project. Nothing is sent anywhere except the model calls you configure.

STEP 02 Deploy on Railway

Open railway.com/deploy/smart-marking-1 and click **Deploy Smart Marking**. Railway creates two services — `web` (the app) and `Postgres` (the database) — plus a small disk for page images. The first build takes two to three minutes.

Your sign-in password is generated for you. To read it: open the `web` service in Railway → **Variables** → `TEACHER_PASSWORD`. You can change it there at any time (the service restarts). To open the app, click the `web` service → **Settings** → **Networking** and follow the public domain (it looks like `web-production-xxxx.up.railway.app`).



The template page on Railway. One click creates the app, the database and the storage.

STEP 03 Sign in

Open the app's address and enter the teacher password from the step above. There is one shared teacher password per school deployment.

Smart Marking

Sign in

Teachers only. Enter the password your school set when it deployed Smart Marking.

Password

Sign in

Smart Marking · self-hosted on Railway



Handwritten scripts marked against your rubric. You check the doubtful ones, then release.

The sign-in page. Teachers only — students never sign in.

STEP 04 Connect a model

Go to **Settings**. Pick a provider, paste its key, then click **Load models from provider** to see exactly which models your key can use. Choose a model marked *reads pages*, click **Test connection**, and **Save**.

Keys are kept per provider (a ✓ on the tile means a key is saved), so you can switch between providers without re-entering them. **Requests per minute** protects free keys with low rate limits — set it to 8 for TokenRouter's free model, 10 for Gemini's free tier.

My models. OpenRouter and TokenRouter host hundreds of models beyond the curated list. Under those two providers a *My models* list lets you add any model id (with a label and whether it reads pages); it then appears in every model picker, here and in the assignment editor. Each id returned by *Load models from provider* gets a + **Add** button.

Smart Marking

ClassesSubmissionsAssignmentsReviewLearningSettingsSign out

AI model

Which model marks your scripts, and the key it uses.

Provider

TokenRouter ✓	OpenRouter ✓	OpenAI	Anthropic	Moonshot (Kimi)	Qwen (Alibaba Model Studio)	Google Gemini
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Free options: Auto Router (openrouter/auto) may pick a text-only model. If you use it, set a vision model under "Different model for reading pages". OpenRouter's :free models cost nothing (~50 requests/day until you've bought \$10 of credits, then 1,000/day) — load models and look for the :free suffix.

Model

Auto Router (picks a model per request) · text on▼

Load models from provider

API key

Where to get a OpenRouter key

Remove OpenRouter key

Different model for reading pages (optional)

leave blank to use the same model

Use a cheap vision model to transcribe, and a stronger one to mark.

Requests per minute

60

0 = no limit. A script needs about 4 requests.

Ask me when confidence is below

0

Settings: provider, model, key. Test connection makes one small text call and one image call.

My models

Model ids you add stay in the Model list for OpenRouter.

None yet.

Model id	Label
exact id, e.g. google/gemini-3.8-flash	optional — what the Model list shows

☒ Reads pages

Add model

My models: your own OpenRouter or TokenRouter model ids, available everywhere a model is picked.

STEP 05

Create an assignment

Go to **Assignments** → **+ New assignment**. Choose the type: *Maths / Science — mark scheme* for structured answers, *Essay — rubric* for open responses, or *Quick*

mark for a short criteria list with no paper.

Give it a title, drop the **question paper** (PDF or photos) and click **Read questions** — the app transcribes every question and part into a table you can correct. Then drop the **mark scheme** and click **Read mark scheme**: each row shows the expected answer and its mark allocation (M1, A1, B1...). Add any notes the marker should know ("ECF applies", "accept any correct method"), check the tables, and **Save**.

Saved assignments go into a bank you can reuse across classes and years.

Smart Marking

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[← Assignments](#)

Quadratic equations — Worksheet 3

Choose the type, add the paper and the mark scheme, then save. Drafts are kept as you go.

2 questions · 3 marks

Type

Maths / Science — mark scheme

Essay — rubric

Quick mark

Mark scheme: marks per question part (M1, A1...). Rubric: bands per criterion for an essay. Quick mark: a short criteria list, no paper.

Title

Quadratic equations — Worksheet 3

Subject

Maths

English

Science

Question paper

Upload the paper and we read the questions and parts; check the table and correct anything.

Drop the question paper here

PDF, JPG, PNG or HEIC — replaces the current pages

Choose files

1 page uploaded

Read questions

LABEL	ID	QUESTION	MARKS
1(a)	1a	Solve $2x + 3 = 11$.	2
1(b)	1b	Hence find x^2 .	1
			3

+ Add question

Mark scheme

One row per question part, with its allocations (M1, A1, B1...). Every question needs a row.

Drop the mark scheme here

PDF, JPG, PNG or HEIC — replaces the current pages

Choose files

1 page uploaded

Read mark scheme

QUESTION # PART	EXPECTED ANSWER	ALLOCATION	MARKS	NOTES
1a	$2x = 8; x = 4$	M11A11	2	Accept any correct method
1(b)	16	B11	1	Accept any correct method
			3	

+ Add row

Notes

Anything the marker should know.

Notes (optional)

Accept any correct method. ECF applies.

Delete student pages after marking

Follow default (on)

Student pages are deleted as soon as a script is done; the marking record keeps the transcription and every mark. Change the default under Settings.

Save

Ready to save.

The assignment editor after Read questions and Read mark scheme. Every cell is editable.

Smart Marking
Classes
Submissions
Assignments
Review
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Sign out

Assignments

Papers, mark schemes and rubrics you have saved. Pick one under Mark a script instead of typing it again.

+ New assignment
Import JSON
Export JSON

TITLE	SUBJECT	SCHEME	CRITERIA	MARKS	TIMES USED	UPDATED			
Quadratic equations — Worksheet 3 Accept any correct method. ECF applies.	Maths	Mark scheme Paper: 1 page	1	1	0	Thu, 17 Sept, 9:34 am	Rename	Duplicate	Delete
Revision Paper	Science	Mark scheme Paper: 13 pages	40	70	0	Thu, 17 Sept, 7:46 am	Rename	Duplicate	Delete

The assignment bank.

STEP 06 Choose the model per assignment

Every assignment follows Settings by default (*Auto — follow Settings*): change the model in Settings and every assignment changes with it. For one assignment that deserves something different — a hard paper for a stronger model, a quick quiz for a cheap one — open it, go to **Model** and pick *Choose a model*. Only providers with a saved key can be selected; choose the model, optionally a different one for reading pages, and **Save**. The assignment bank shows the choice as a small caption.

Removing a provider's key while assignments still use it asks you to confirm first — those assignments would fail to mark until you point them at another model.

Model

Which model marks this assignment. Auto follows Settings, so changing Settings changes it too.

Auto — follow Settings

Choose a model

Provider

TokenRouter	OpenRouter	OpenAI	Anthropic	Moonshot (Kimi)	Qwen (Alibaba Model Studio)	Google Gemini
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Model

Different model for reading pages (optional)

GLM 5.3 Flash · reads pages

leave blank to use the same model

Use a cheap vision model to transcribe, and a stronger one to mark.

The Model section of an assignment: Auto, or a provider and model of your own.

STEP 07 Create a class and load the classlist

Go to **Classes** → **New class** and name it the way your school does ("4E2 Mathematics"). Every class gets a 4-character code — this one is **4KEF** — that students will type.

On the **Students** tab, upload a CSV with two columns, **name** and **reg_no** (register number). The preview flags duplicate or missing numbers before anything is saved; click **Confirm classlist** when it looks right.

Then click **Copy link** and paste it into Google Classroom (or wherever your students look). The link opens the app with the class code already filled in.

Classes

Each class has a code students

No classes yet

Create a class, upload its classlist, the

[+ New class](#)

New class

Class name

[Cancel](#)[Create class](#)

Creating a class.

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Classes

Submissions

Assignments

Review

Learning

Settings

Sign out

← Classes

4E2 Mathematics

Class code 4KEF · 0 students · 0 open

Copy link

Students

Assignments

Settings

Choose CSV

#	NAME	ISSUE
1	Tan Wei Ling	
2	Muhammad Danish Bin Rosli	
3	Priya Nair	
4	Lim Jun Hao	

4 students. Looks good.

Confirm classlist

Cancel

CSV preview — issues are shown per row before you confirm.

Smart Marking

ClassesSubmissionsAssignmentsReviewLearningSettingsSign out

← Classes

4E2 Mathematics

Class code 4KEF · 4 students · 0 open

Copy link

Students

Assignments

Settings

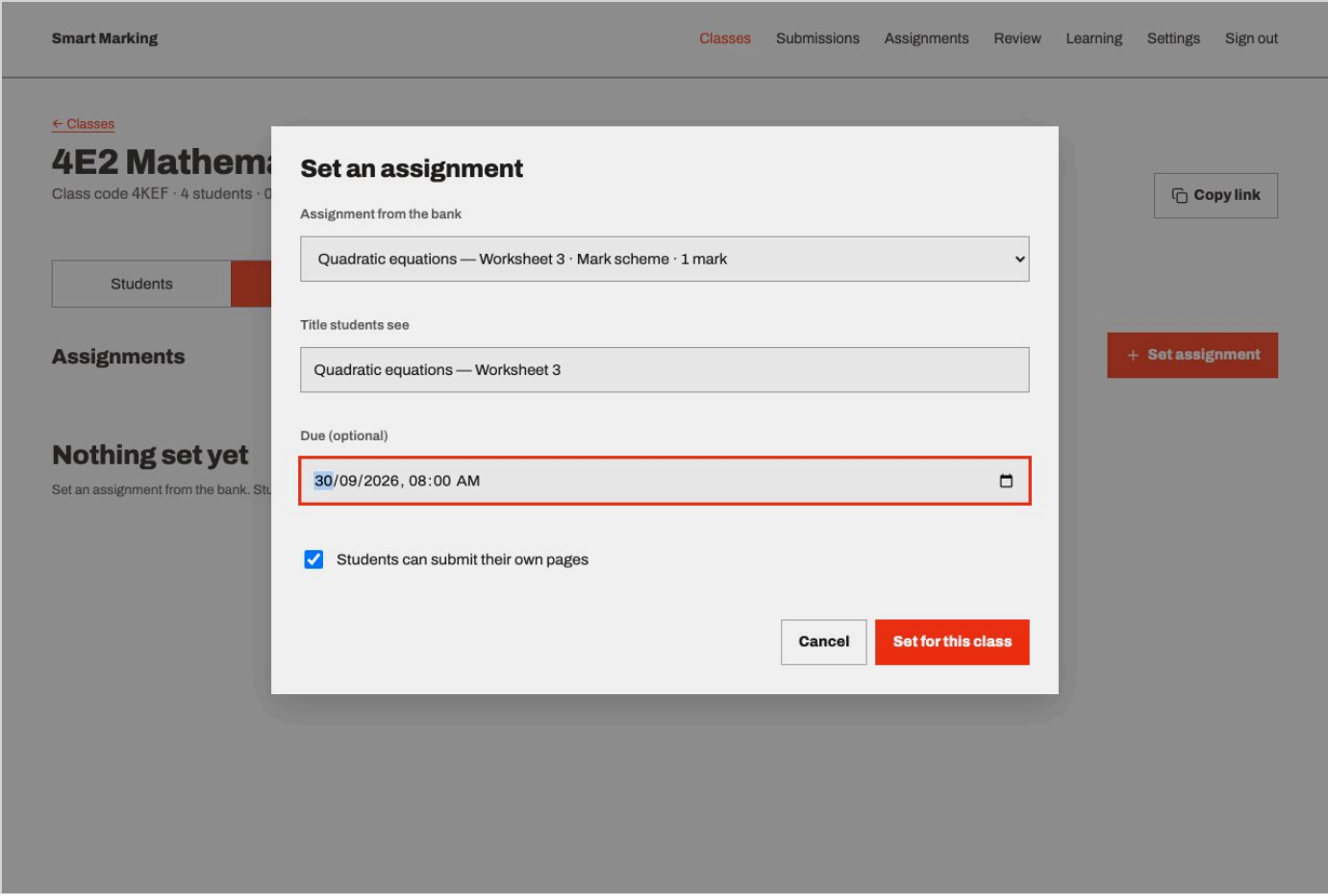
Replace classlist

#	NAME	SUBMISSIONS	LAST SEEN
1	Tan Wei Ling	0	—
2	Muhammad Danish Bin Rosli	0	—
3	Priya Nair	0	—
4	Lim Jun Hao	0	—

The classlist, the class code and the Copy link button.

STEP 08 Set the assignment for the class

On the class's **Assignments** tab click **Set assignment**, pick one from the bank, add a due date and click **Set for this class**. It starts as a *Draft*, invisible to students. Click **Open** when you want students to hand in.



Setting an assignment from the bank.

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ClassesSubmissionsAssignmentsReviewLearningSettingsSign out

[← Classes](#)

4E2 Mathematics

Class code 4KEF · 4 students · 1 open

Copy link

Students

Assignments

Settings

Assignments

+ Set assignment

TITLE	STATUS	DUE	HAND-INS	
Quadratic equations — Worksheet 3 Mark scheme	■ Open	Wed, 30 Sept, 8:00 am	0	Back to draftDelete

Opened — students can now see it.

STEP 09 Students hand in from their phones

A student opens the class link, types their register number and confirms their name. No account, no password: anyone with the class code and a number can see that student's work — the same trade-off as a paper script left on a desk.

Hand in takes photos with the camera or from the gallery. Pages can be re-ordered or deleted before handing in; photos are shrunk on the phone before upload, so it works on a weak signal. Up to 20 pages per script. Once handed in, the student sees a calm "Handed in" note — marking starts straight away in the background.

Enter your number

4KEF-

Continue

Your teacher shared a link and your register number is on the class list.

Is this you?

Are you Tan Wei Ling, #1 of 4E2 Mathematics?

Yes, that's me

Not me

Smart Marking Tan Wei Ling · #1 **Not me?**

4E2 Mathematics

Quadratic equations — Worksheet 3

Due Wed, 30 Sept, 8:00 am

To hand in

Enter your number.

Confirm it's you.

Home: what to hand in.

Smart Marking Tan Wei Ling · #1 **Not me?**

Quadratic equations — Worksheet 3

Due Wed, 30 Sept, 8:00 am

Take photo

Choose from gallery

One photo per page, in order — up to 20 pages.
Photos are shrunk on your phone before they're sent.

1

script.png

Hand in 1 page

Smart Marking Tan Wei Ling · #1 **Not me?**

Quadratic equations — Worksheet 3

Handed in Thu, 17 Sept, 9:35 am

Your teacher will see it shortly. You'll find your feedback here once it's ready.

Back to assignments

Smart Marking Tan Wei Ling · #1 **Not me?**

Quadratic equations — Worksheet 3

Handed in Thu, 17 Sept, 9:35 am.

Marking usually takes a day. We'll show your feedback here.

Back to assignments

Pages in order, ready to hand in.

Done.

Waiting for the teacher.

STEP 10 Marking, and the parts that need you

The assignment page shows the roster: who has handed in, what is being marked, and what is ready. You can also **Upload pages** for a student yourself (scanned booklets)

and **Remove hand-in** so a student can redo it.

Each script is read by one model, marked against the scheme by a second pass, and checked by an independent reviewer. Anything illegible, not covered by the scheme, or where marker and reviewer disagree goes to **Review** with the reason. There you tick the allocations the student earned (or pick the band for an essay) and give a one-line reason — that reason feeds the nightly learning step, so the marker gets closer to your judgement over time.

Click a student's name to see the per-part marks, the transcription and the justification for each mark.

Smart Marking

ClassesSubmissionsAssignmentsReviewLearningSettingsSign out

[← 4E2 Mathematics](#)

Quadratic equations — Worksheet 3

■ Open Due Wed, 30 Sept, 8:00 am · 4 students

Download marks CSV

Download marking records

Release feedback

3 Not handed in	0 Handed in	0 Marking	0 Needs you	1 Ready
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#	NAME	PAGES	HANDED IN	STATUS	TOTAL	
1	Tan Wei Ling	1	Thu, 17 Sept, 9:35 am	■ Ready	3 / 3	Remove hand-in →
2	Muhammad Danish Bin Rosli	—	—	■ Not handed in	—	Upload pages
3	Priya Nair	—	—	■ Not handed in	—	Upload pages
4	Lim Jun Hao	—	—	■ Not handed in	—	Upload pages

The roster after marking: 1 ready, 0 needing you.

← Submissions

#1 Tan Wei Ling

Quadratic equations — Worksheet 3 · Maths · Accept any correct method. ECF applies. · uploaded Thu, 17 Sept, 9:35 am · 1 page · marked Thu, 17 Sept, 9:36 am by openrouter/auto

Download marking record

TOTAL
3 / 3

Done

Pages

Pages deleted after marking

Student pages are deleted as soon as a script is done; the marking record keeps the transcription and every mark.

Marks by question part

QUESTION & PART	SCHEME ANSWER	STUDENT'S ANSWER	JUSTIFICATION	AWARDED
1(a)	$2x = 8; x = 4$	$x = 4$	The student correctly rearranged to " $2x = 8$ "	M1 ✓
Solve $2x + 3 = 11$.	4	Workings: $2x = 8$	and obtained " $x = 4$ ".	A1 ✓
	[M1 1 · A1 1]			2 / 2
1(b)	16	$x^2 = 16$	The student correctly found $x^2 = 16$.	B1 ✓
Hence find x^2 .	[B1 1]			1 / 1

Feedback report

Great work on this short algebra exercise! You correctly solved both parts by rearranging the equations to find the unknown values. Keep practising these types of problems to build even more fluency.

WHAT YOU DID WELL

- You accurately rearranged the equation in 1a to get $2x = 8$, showing a clear understanding of inverse operations.
- You correctly solved for x in 1a, finding $x = 4$.
- You confidently computed $x^2 = 16$ in 1b, applying the square correctly to your answer.

QUESTION BY QUESTION

1a Excellent! You correctly moved the 3 to the other side by subtracting it from 11, giving $2x = 8$. Then you divided by 2 to find $x = 4$. Your working is clear and accurate. *Try next: Practise similar two-step equations such as $3x + 5 = 20$ to make the process even more automatic.*

1b Well done! Since $x = 4$, you correctly calculated $x^2 = 16$. This shows you know what squaring a number means. *Try next: Try evaluating squares and cubes of negative numbers as a next challenge (for example, if $x = -4$, what is x^2 ?).*

WORK ON NEXT

- Practise solving two-step equations with positive and negative numbers.
- Check your solutions by substituting them back into the original equation.
- Explore squares and cubes of small integers, including negatives.
- Write a one-line check after each solution to catch small mistakes.

NEXT STEPS

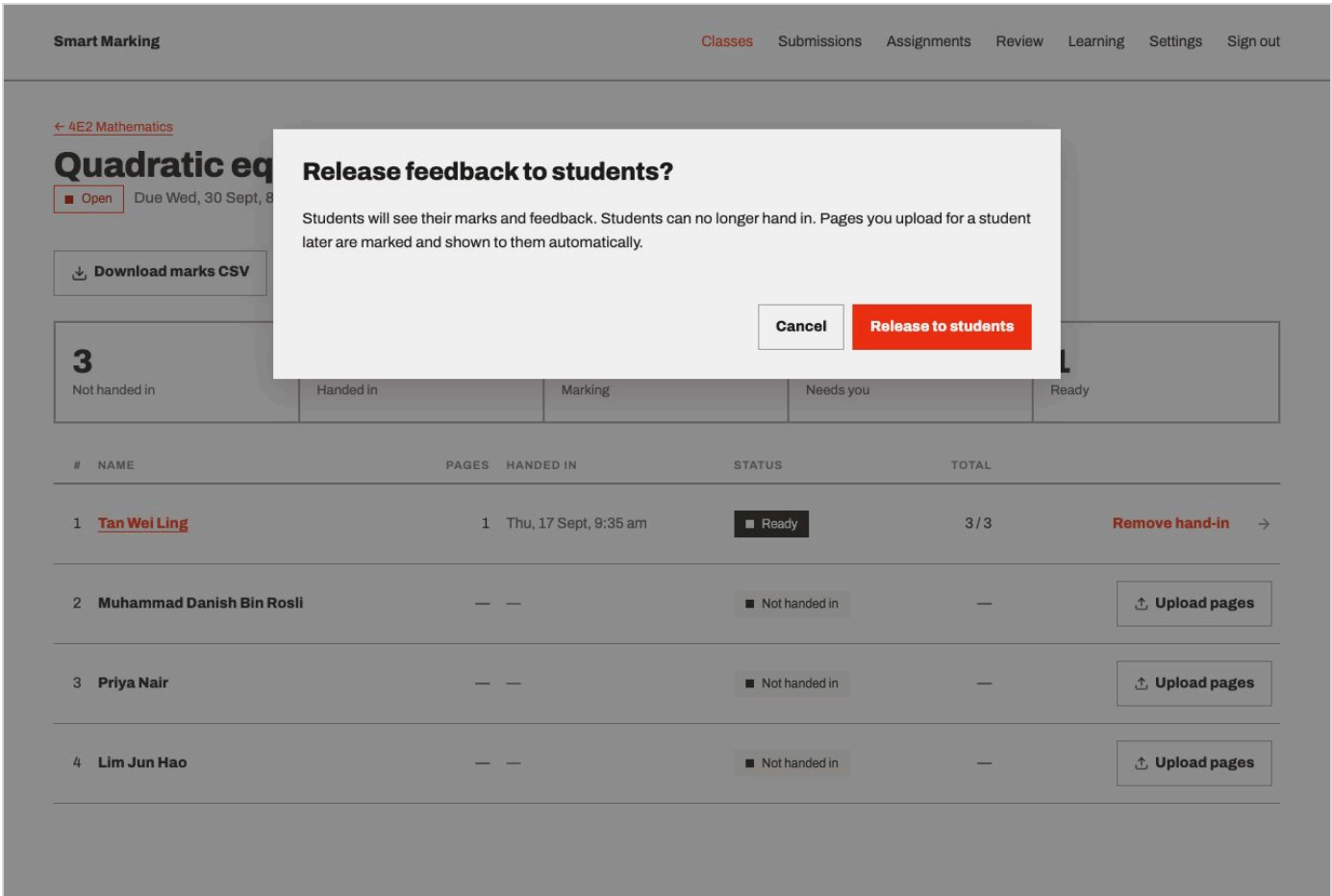
- Complete 5 more equations like $2x + 3 = 15$ and show each step.
- Use substitution to verify that $x = 4$ solves the original equation $2x + 3 = 11$.
- Practise finding x^2 and x^3 for values like 2, 3, 5, and -2.

A marked script, part by part.

STEP 11 Release feedback

Until you release, students see "Marked — your teacher is checking". **Release feedback** is one action for the whole class and needs every "Needs you" part cleared first. After release students see their total, a short summary, what they did well, each question with a comment and a "Try next", and what to work on.

Releasing closes student hand-ins for that assignment; pages you upload for a student later are still marked and shown automatically.



The release confirmation.

Smart Marking

Tan Wei Ling · #1

Not me?

[Back to assignments](#)

Quadratic equations — Worksheet 3

Handed in Thu, 17 Sept, 9:35 am

3

/ 3

Great work on this short algebra exercise! You correctly solved both parts by rearranging the equations to find the unknown values. Keep practising these types of problems to build even more fluency.

What you did well

- You accurately rearranged the equation in 1a to get $2x = 8$, showing a clear understanding of inverse operations.
- You correctly solved for x in 1a, finding $x = 4$.
- You confidently computed $x^2 = 16$ in 1b,

What you did well

- You accurately rearranged the equation in 1a to get $2x = 8$, showing a clear understanding of inverse operations.
- You correctly solved for x in 1a, finding $x = 4$.
- You confidently computed $x^2 = 16$ in 1b,

Question by question

1(a)

2 / 2 ^

Excellent! You correctly moved the 3 to the other side by subtracting it from 11, giving $2x = 8$. Then you divided by 2 to find $x = 4$. Your working is clear and accurate.

Try next: Practise similar two-step equations such as $3x + 5 = 20$ to make the process even more automatic.

WHAT WE READ FROM YOUR PAGE

$x = 4$

What the student sees.

A question expanded.

STEP 12 Insights: where the class lost marks

Once the first script is marked, the assignment's **Insights** tab shows the class as a whole: marks by part with the weakest highlighted, the allocations most often lost, a score distribution, and which students are struggling on which parts.

When the whole set has been marked, and again when you release, the app asks the model for a short narrative written for you, not for students — a summary, strengths, gaps and recommended next steps. Student names never reach the model: it sees register numbers and the app joins the names back. **Regenerate** refreshes the narrative after more scripts come in; **Download PDF** gives a one-file report for a department meeting.

Smart Marking

ClassesSubmissionsAssignmentsReviewLearningSettingsSign out

[← 4E2 Mathematics](#)

Quadratic equations — Worksheet 3

Released

Due Wed, 30 Sept, 8:00 am · 4 students

Download marks CSV

Download marking records

Roster

Insights

1 of 4 marked · mean 3 / 3

Generated Thu, 17 Sept, 8:51 pm · OpenRouter openrouter/auto

Regenerate

Download PDF

Marks by part

Mean % of the marks available, in scheme order. The weakest parts are amber.

1(a)		100%
1(b)		100%

Summary

Only 1 of 4 scripts is marked (reg 1), and that script earned 3/3: full marks on 1(a) and 1(b), with no allocation lost. The mean is therefore 3, but this is a single-script sample; it is not yet evidence of class-wide mastery.

Strengths

- 1(a): M1/A1 both earned — reg 1 rearranged $2x = 8$ and solved $x = 4$ without losing the coefficient.
- 1(b): B1 earned — reg 1 produced $x^2 = 16$, completing the worksheet's credited step.

Recommended next steps

- Mark or collect the other three scripts before planning remediation** (1(a), 1(b))
Only reg 1 has marks. Use the next five minutes to mark the remaining submitted scripts; otherwise the full marks on 1(a)/1(b) should not be treated as a class pattern.
- Model the equality-preserving line for 1(a)** (1(a))
Display reg 1's two-line method ($2x = 8$, then $x = 4$) and ask students to state which inverse operation is used at each line. This reinforces the M1 step before the A1 answer.
- Check whether 1(b) should end at $x = \pm 4$** (1(b))
The marked script stops at $x^2 = 16$, which earned B1. If the intended outcome is solving the quadratic, add a follow-up prompt such as 'now give both solutions' and mark the $\pm$ roots as a separate allocation.
- Set one same-structure starter for the whole class** (1(a), 1(b))
Give four short items in the same form as 1(a) and 1(b) — solving $ax = b$ and finding $x^2 = c$ — as a 5-minute entry task. This tests whether the remaining students can reproduce the M1/A1/B1 steps without new teaching.

Score distribution

How many scripts scored in each band, out of 3.

0–0		0
1–1		0
2–2		0
3–3		1

The Insights tab: numbers first, then the narrative and the students to support.

STEP 13

Telegram notifications (optional)

Get a message when scripts come in and when marking finishes, plus a daily report. In Telegram, message **@BotFather**, send `/newbot`, and copy the token it gives you. In

Settings → Notifications paste the token and **Save**, then open your new bot in Telegram and press **Start** — within a few seconds the status changes to *Linked* ✓. **Test connection** sends a test message.

You get an instant message for new hand-ins (batched) and one when a class set finishes marking, each with links back to the app. The daily report arrives at the time you set — 07:00 Singapore time by default — and only on days something happened. Turn instant messages off to keep just the daily report. One chat per deployment: a department group works well — add the bot to the group and press `/start` there.

Notifications

Telegram messages when scripts come in and when marking finishes, plus a daily report.

Telegram bot token

Paste the token from @BotFather

Not linked — save the token, then open your bot in Telegram and press /start

App URL

auto-detected on Railway

Where the links in your Telegram messages point.

☒ Send instant messages (hand-ins, marking finished)

Turn this off to get the daily report only.

Daily report at

07:00 AM

⌚

Time zone

Asia/Singapore

▼

Test connection

Save

Settings → Notifications: token, link status, instant messages, daily report time and time zone.

STEP 14 Marks and records

From the assignment page, **Download marks CSV** gives one row per student with a column per question part, and **Download marking records** gives a Word document per student — mark scheme row by row, the student's answer, the awarded mark and a blank "Teacher's mark" column for moderation — plus a markbook spreadsheet.

STEP 15 Good to know

Student pages are deleted as soon as a script is done (marked with nothing to check, or the last part resolved); the marking record keeps the transcription and every mark. Turn this off under Settings or per assignment if you want to keep the images.

Cost. A script needs about four model calls. On a paid provider that is typically well under one cent per script for a small model; on a free tier it is free within the daily limit.

Changing the password or the secret. The password is the `TEACHER_PASSWORD` variable on Railway. Leave `SECRET_KEY` alone — it encrypts the stored API keys; changing it means re-entering them.

Telegram and Insights are optional. Nothing is sent to Telegram until you link a bot, and Insights only use the model you already configured — no extra keys.

Updates. Redeploying the `web` service picks up the latest version from GitHub; the database and stored keys are kept.

Source, issues and the full README: github.com/twahidin/smart-marking-system.